



**Gokaraju Rangaraju Institute of Engineering and Technology
(Autonomous)**

Department of Data Science

**Mini Project Work
(CSE(DS))**

Review 1: Abstract Presentation

Batch No - 14

Presented by:

- 1. Ms. Kanchari Neeharika (24241A3223)**
- 2. Ms. Hima Sree Krothapally (24241A3228)**
- 3. Ms. Polaju Ananya (24241A3247)**

Under the Guidance of:

**V. Y. Bharadwaj
Asst. Professor
Department of Data Science**

REBID AI

Proposed Solution

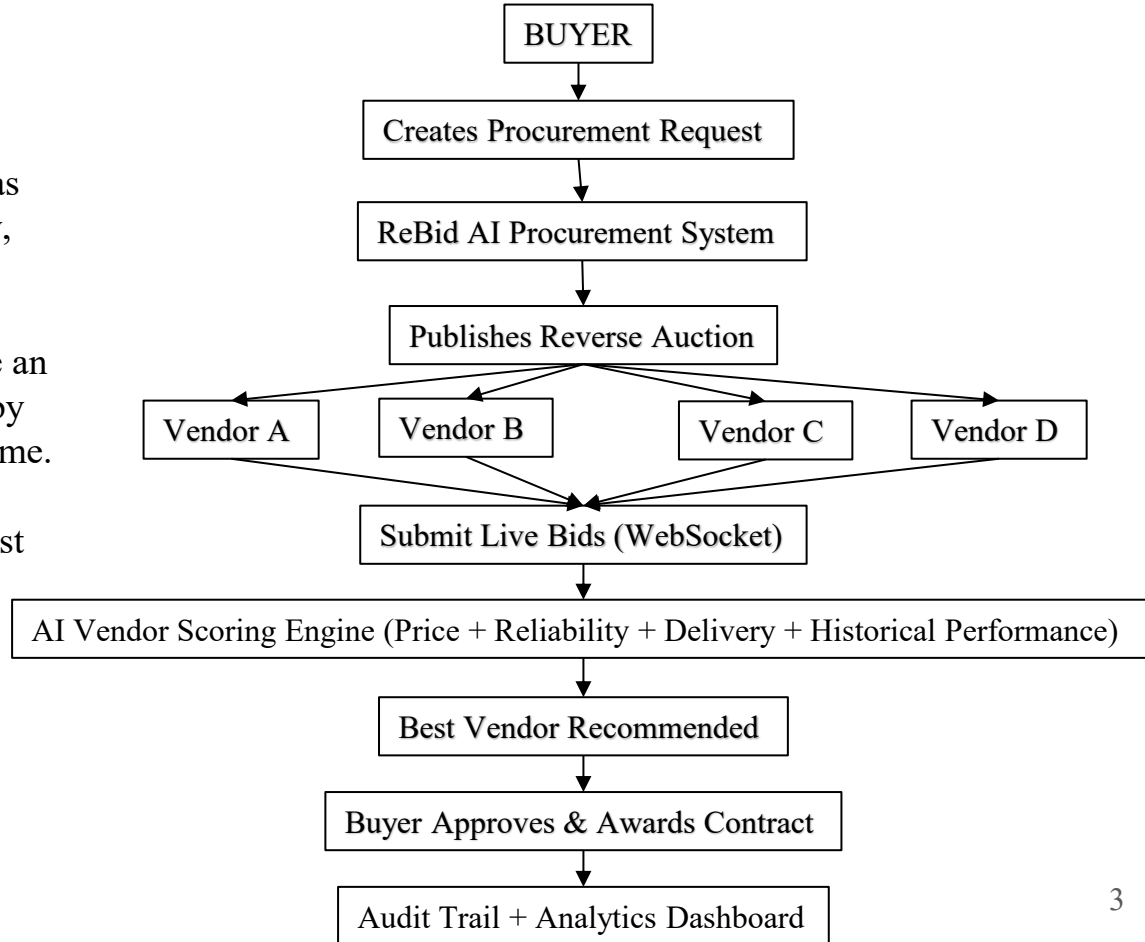
Problem

Procurement that is traditionally carried out tends to favor the bid that offers the lowest prices while failing to consider such factors as reliability of the vendor, performance history, and delivery record.

Problem Solution

ReBid AI uses artificial intelligence to create an intelligent reverse auctioning system, whereby vendors compete to lower their bids in real time. The system does not base its decision on the price of the bid only but recommends the most suitable vendor according to the following parameters:

- Bid Price
- Vendor Reliability
- Delivery Performance
- Performance History



Modules and Data Flow

Buyer Portal

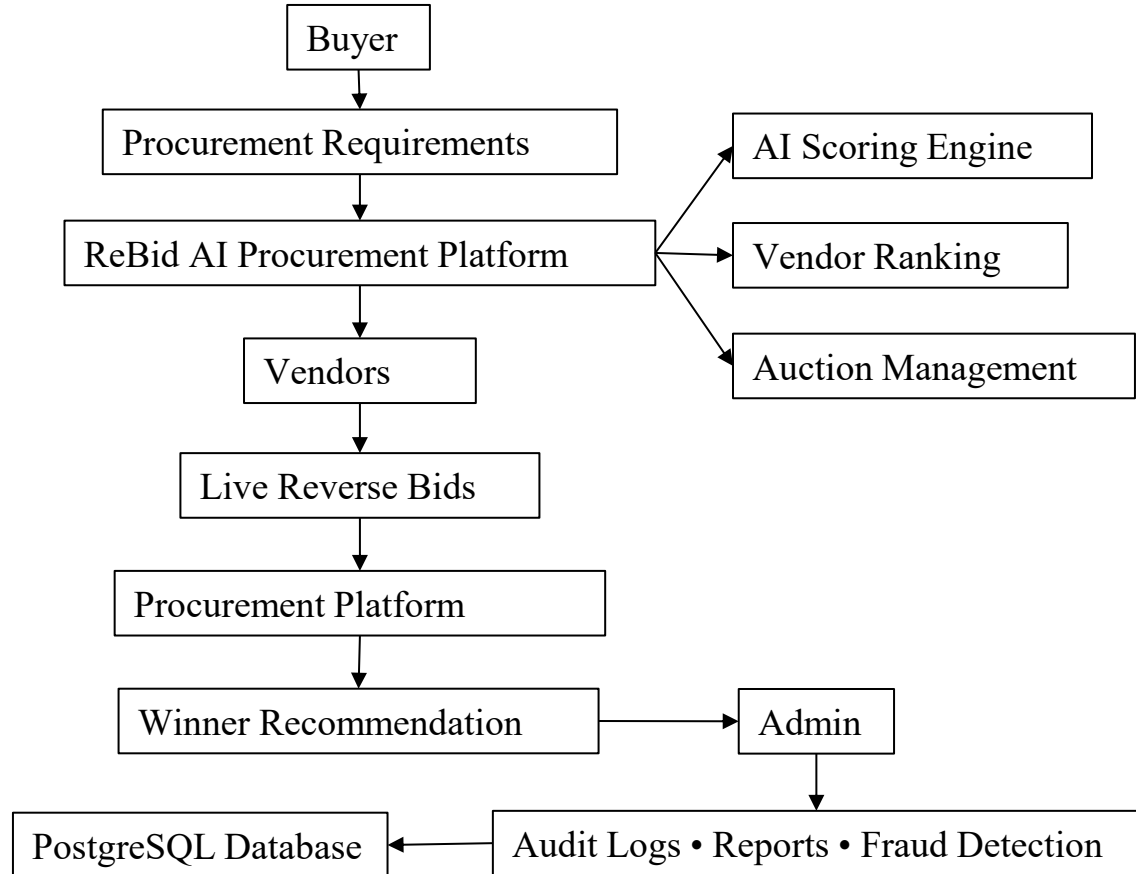
- Create auction
- Compare vendors
- Award contract
- View analytics

Vendor Portal

- Participate in auctions
- Track rankings
- View reputation score
- Receive notifications

Admin Portal

- Verify vendors
- Detect fraud
- Manage users
- Generate reports

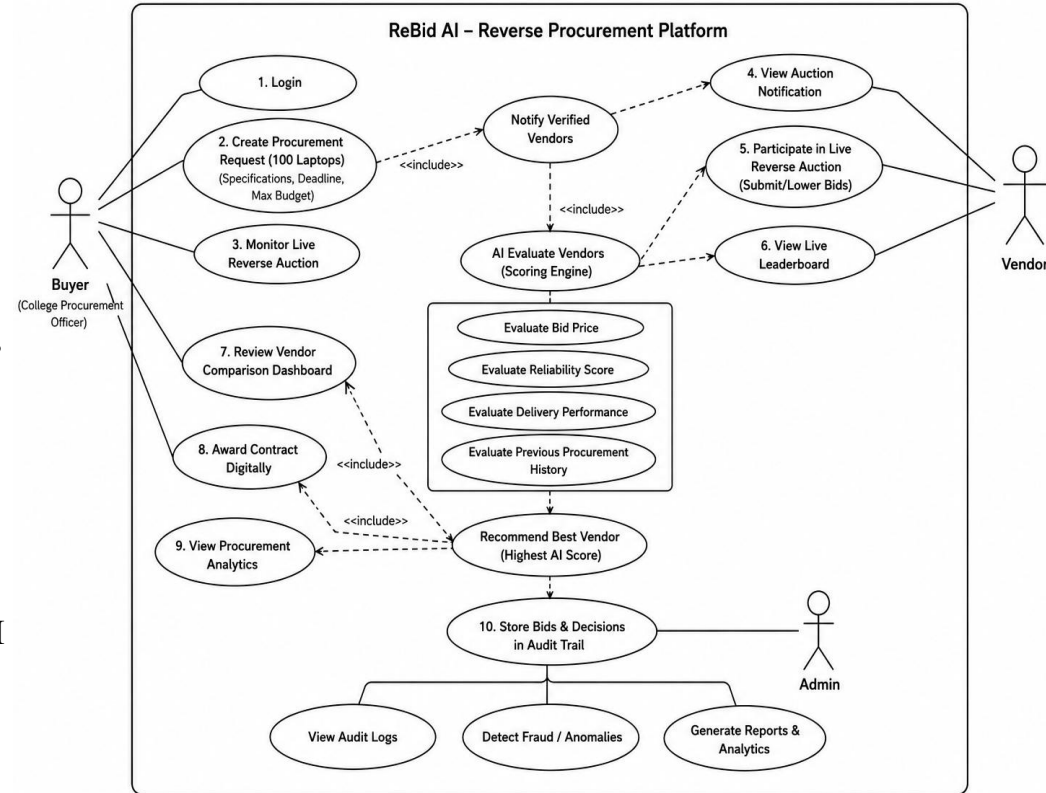


Use case

Use Case: Procurement of 100 Laptop Computers

Actor: Buyer (Procurement Officer in College)

- Buyer signs into ReBid AI.
- Buyer initiates a procurement process for 100 laptop computers along with specifications, deadlines, and maximum budget.
- Verification of vendors results in immediate notifications to them.
- Vendor conducts real-time reverse auction and keeps bidding down.
- AI analyzes every vendor on the basis of:
 - Bid Cost
 - Reliability Rating
 - Delivery Performance
 - Previous Procurement Experience
- Recommendation is made for vendor with highest AI rating rather than lowest bid cost.
- Buyer views the comparison dashboard and makes procurement digitally.
- Every bid and procurement decision is recorded in audit trail.



Base paper

Title: Incentive-Driven Reverse Auctions for Obtaining Transportation Services with Cost Variation and Delivery Uncertainty

Authors: X. Chen, Y. Wang, X. Zhao

Publication: PLOS ONE, 2018

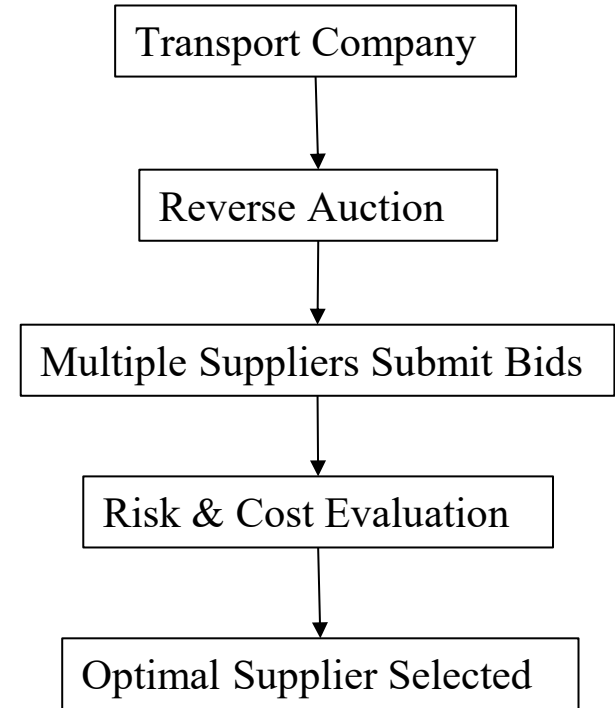
Problem Statement:

Reverse auctions are a popular procurement method among many companies in logistics. However, the major problem with this procedure is that the lowest bidder is always selected by the mechanism, and this leads to the following issues:

Delivery problem:

- Untrustworthy suppliers
- Undisclosed risks
- Rise in the issues connected with procurement

The authors suggest an incentive-based reverse auction method that considers both costs and delivery risks with respect to choosing a supplier.



Base paper Methodology

Techniques:

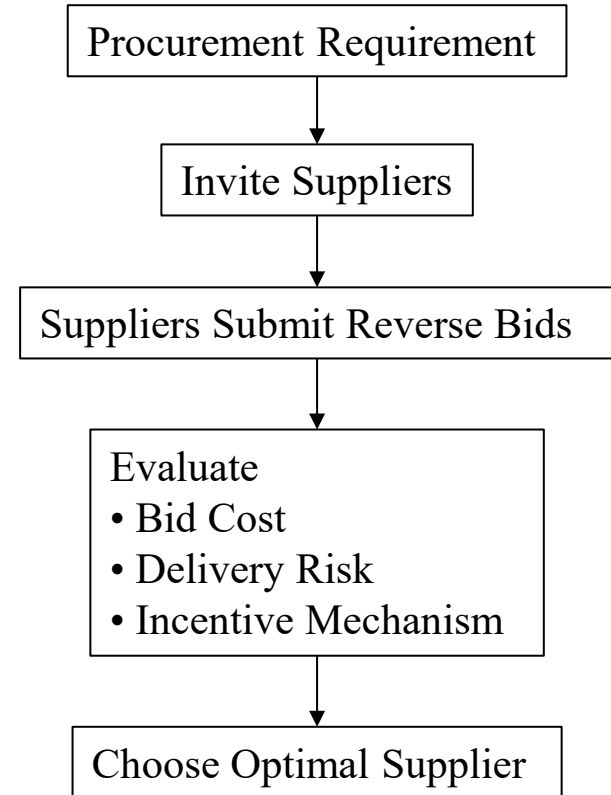
- Reverse Auction
- Optimization
- Risk Analysis of Delivery
- Incentive Model
- Cost Minimization

Advantage:

- Decreases procurement costs
- Promotes supplier competition
- Takes into account delivery risks
- Enhances logistics procurement

Limitations:

- Primarily developed for purposes of transportation/logistics procurement.
- Relies on a mathematical optimization model and not deployable software
- Not a real-time online auction
- Does not have AI-driven vendor selection
- Lacks audit trails
- No analytics dashboard available
- No fraud detection capabilities
- No live notifications
- No vendor reputation system



Comparison with Base paper

Base Paper

Title: Incentive-Based Reverse Auctions for Transportation Service Procurement with Cost Uncertainty and Delivery Risk (PLOS ONE, 2018)

Student Proposal

Title: ReBid AI : AI-Powered Reverse Procurement Platform with Intelligent Vendor Scoring

Base Paper (2018)	Student Proposal – ReBid AI
Focused on transportation/logistics procurement .	Supports any enterprise/B2B procurement such as IT equipment, office supplies, raw materials, and services.
Used a mathematical incentive-based reverse auction model .	Implements a real-time web-based reverse auction platform using FastAPI, React, and WebSockets.
Evaluated suppliers using bid cost and delivery risk .	Evaluates vendors using Bid Price + Reliability Score + Delivery Performance + Procurement History .
Recommended suppliers using optimization algorithms .	Uses AI/ML models (XGBoost/LightGBM) to intelligently rank and recommend vendors.
Research focused mainly on auction theory and optimization .	Provides a complete deployable procurement solution with Buyer, Vendor, and Admin portals.
Did not include live bidding interface .	Supports real-time reverse auctions with live bid updates and leaderboards.
No audit or transparency features.	Includes tamper-evident audit trail , digital contract award, and procurement history.
No analytics or reporting.	Provides procurement analytics dashboards , spend analysis, and vendor performance reports.
No fraud detection mechanism.	Includes AI-based fraud and anomaly detection for suspicious bidding behavior.
No enterprise security implementation.	Uses JWT authentication , role-based access control, and secure APIs.

Literature Review

1. Incentive-Based Reverse Auctions for Transportation Service Procurement with Cost Uncertainty and Delivery Risk (2018)

- Proposes an incentive-based reverse auction considering both procurement cost and delivery risk.
- Selects suppliers using optimization instead of lowest price alone.
- Forms the foundation for ReBid AI's reverse auction mechanism.

2. Online Reverse Auctions: Issues, Themes, and Prospects for the Future (Jap, 2002)

- Explores the adoption of online reverse auctions in B2B procurement.
- Shows that reverse auctions improve transparency and reduce procurement costs.
- Highlights challenges in supplier relationships and bidding strategies.

3. Procurement Auction Design: A Review (Beil, 2010)

- Reviews different procurement auction mechanisms and their effectiveness.
- Discusses multi-attribute supplier evaluation beyond price.
- Recommends optimizing auction rules for better supplier selection.

4. Auction Theory (Vijay Krishna, 2010)

- Explains the principles and strategies behind auction mechanisms.
- Compares forward and reverse auctions with bidder behavior.
- Provides theoretical support for competitive bidding systems.

5. Supplier Selection Using Multiple Criteria Decision Making: A Review (Chai et al., 2013)

- Reviews supplier selection methods using multiple decision criteria.
- Demonstrates that quality, reliability, and delivery should complement price.
- Supports AI-based vendor evaluation in ReBid AI.

Literature Review

6. A Review of Machine Learning Applications in Supply Chain Management (Carbonneau et al., 2008)

- Surveys machine learning applications in supply chain operations.
- Shows ML improves forecasting and procurement decisions.
- Inspires AI-driven vendor recommendation.

7. Predictive Manufacturing Systems Using Machine Learning (Wuest et al., 2016)

- Uses predictive analytics for operational decision-making.
- Demonstrates improved efficiency through ML models.
- Relevant for procurement analytics and performance prediction.

8. Supplier Performance Evaluation Framework (Kwon et al., 2010)

- Evaluates suppliers based on historical performance indicators.
- Highlights the importance of reliability and delivery records.
- Supports the reputation scoring feature.

9. Intelligent Procurement Using Artificial Intelligence (Li et al., 2020)

- Introduces AI-based procurement automation.
- Demonstrates improved efficiency and decision support.
- Encourages AI integration into procurement platforms.

10. Fraud Detection Using Machine Learning Techniques (Ahmed et al., 2016)

- Applies anomaly detection to identify suspicious activities.
- Improves fraud prevention in online transactions.
- Supports ReBid AI's bid collusion detection module.

Literature review

11. JWT-Based Authentication for Secure Web Applications (Sandhu et al., 2021)

- Describes secure authentication using JSON Web Tokens.
- Implements role-based access control for web systems.
- Forms the security foundation of ReBid AI.

12. Blockchain Technologies in Supply Chain and Procurement (Pournader et al., 2020)

- Explores blockchain for procurement transparency.
- Improves traceability and auditability.
- Inspires tamper-evident audit trail concepts.

13. Digital Procurement: The Future of Sourcing (Gartner, 2023)

- Explains how AI transforms enterprise procurement.
- Shows automation reduces procurement cycle time.
- Supports the business relevance of ReBid AI.

14. Global Chief Procurement Officer Survey (Deloitte, 2023)

- Studies AI adoption in procurement organizations.
- Reports increased efficiency through analytics and automation.
- Highlights the growing demand for intelligent procurement systems.

15. Multi-Attribute Reverse Auction Mechanisms for Supplier Selection (Talluri et al., 2007)

- Combines price, quality, and delivery metrics in reverse auctions.
- Improves supplier selection compared to lowest-price bidding.
- Directly supports ReBid AI's AI-based vendor scoring model.

Literature Review Summary

- Reviewed 15 research reports regarding reverse auction, AI-enabled acquisition, supplier selection, lying detection, and buying analytics.
- Majority of research finds that reverse auction lead to lesser procurement costs due to competitive bidding process.
- Prior research proposes to opt for multi-criteria evaluation of suppliers instead of going for the lowest bid, i.e., price, quality, dependability, etc.
- Machine Learning as well as AI techniques enhance efficiency of supplier rating, purchasing and forecasting.
- Auctions provide increased openness and fairness and high degree of suppliers' involvement.
- Fraud detection allows to identify collusion, suspicious bidding and supplier actions.
- Purchasing analytics gives opportunity to make data-based buying decisions.
- Authentication and activity tracking provide safety and clarity.
- Most of proposals rely upon particular aspects yet do not give complete procurement solution.
- ReBid AI addresses all of the problems and combines real-time auction, AI-enabled supplier rating, fraud detection, purchasing dashboards, authentication and activity tracking in one intelligent acquiring system.

Literature Review Summary

Source	Research Objective / Question	Methodology	Key Findings	Relevance to Project	Techniques	Accuracy / Performance Measures	Advantages	Disadvantages
Chen et al. (2018)	Improve transportation procurement	Incentive-based optimization	Cost & risk improve supplier selection	Foundation for reverse auction	Reverse Auction, Optimization	Lower procurement cost	Multi-factor evaluation	Limited to logistics
Kagel & Levin (2016)	Study bidder behavior	Experimental auctions	Competition reduces prices	Auction strategy	Auction Theory	Higher bidding efficiency	Fair and competitive bidding	No AI support
Jap (2002)	Improve B2B procurement	Empirical analysis	Online auctions increase transparency	Digital procurement	E-Procurement	Faster procurement	Transparent procurement	Price-centric approach
Beil (2010)	Review procurement auctions	Literature survey	Multi-attribute evaluation is effective	Vendor selection	Multi-Criteria Decision Making	Better decision quality	Comprehensive evaluation	Mostly theoretical
Talluri et al. (2007)	Multi-criteria supplier selection	Optimization model	Quality improves procurement	AI scoring criteria	Optimization	Improved supplier selection	Considers multiple factors	Complex implementation

Literature Review Summary

Source	Research Objective / Question	Methodology	Key Findings	Relevance to Project	Techniques	Accuracy / Performance Measures	Advantages	Disadvantages
Kwon et al. (2010)	Measure supplier performance	Performance analysis	Historical data predicts reliability	Vendor reputation	Performance Metrics	Better ranking accuracy	Reliable vendor assessment	Requires historical data
Chai et al. (2013)	Review supplier selection methods	Systematic review	Hybrid methods perform better	AI vendor scoring	AHP, TOPSIS	Higher selection accuracy	Flexible decision-making	High computational cost
Carbonneau et al. (2008)	ML in supply chains	Machine learning survey	ML improves prediction	AI recommendation	Machine Learning	Better prediction accuracy	Intelligent decisions	Data dependent
Wuest et al. (2016)	Predictive analytics	ML models	Better operational decisions	Procurement analytics	Predictive Analytics	Improved forecasting	Data-driven insights	Needs quality datasets
Li et al. (2020)	Intelligent procurement	AI framework	AI automates procurement	Smart procurement	AI Decision Support	Faster decision-making	Automation	Complex deployment

Literature Review Summary

Source	Research Objective / Question	Methodology	Key Findings	Relevance to Project	Techniques	Accuracy / Performance Measures	Advantages	Disadvantages
Ahmed et al. (2016)	Detect fraud	Anomaly detection	Fraud detected early	Fraud detection module	Anomaly Detection	High detection rate	Improves security	Possible false alarms
Sandhu et al. (2021)	Secure authentication	JWT implementation	Secure role-based access	Platform security	JWT, RBAC	Low authentication overhead	Strong security	Token management needed
Pournader et al. (2020)	Transparent procurement	Blockchain framework	Improves traceability	Audit trail concept	Blockchain	Better transparency	Tamper resistance	Higher implementation cost
Gartner (2023)	Digital procurement trends	Industry report	AI accelerates procurement	Industry relevance	AI, Analytics	Reduced procurement time	Practical insights	Not research-focused
Deloitte (2023)	AI adoption in procurement	Global survey	AI improves efficiency	Future enhancement	AI, Business Analytics	Increased operational efficiency	Industry validation	Survey-based findings

Research Gaps

- **Price-Based Vendor Selection:** In reverse auctions, only the pricing aspect of the vendors is taken into consideration.
- **Absence of AI Models in Reverse Auctions:** Very few research works contain any studies regarding the use of artificial intelligence/for intelligent vendor selection.
- **Lack of Full-Scale Auctioning System:** This research work is theoretical and lacks real implementation.
- **No Multi-Faceted Evaluation:** There is a lack of multi-faceted evaluation of the vendors by using pricing, delivery, reliability, and procurement history.
- **Weak Bid Fraud Detection:** Most of the procurement systems cannot identify fraud in terms of collusion and irregular bidding.
- **Weak Decision-Making Assistance:** The available solutions provide little in terms of procurement intelligence.
- **Non-Transparency:** The credibility of procurement decisions cannot be traced due to lack of any system that permits the tracking of each transaction.
- **Weak Access Control System:** In most of the technologies, secure access control systems, and authentication features are lacking.
- **No Solution that Contents All the Problems:** Previous studies addressed some issues, but they focused either on reverse auctioning, AI, or supplier selection; neither of them provides a comprehensive approach.
- **Inability to deal with Multiple Procurement Types:** The majority of researchers have limited their works to industries like logistics or transport rather than procuring multiple categories including goods and services.

Problem statement

Traditional procurement processes are characterized with manual selection of vendors, or the allocation of procurement contracts relying mostly on the principle of selecting the lowest bidder, while ignoring such important aspects as vendor reliability, delivery punctuality and past performance of vendor. Such approaches lead to inefficiencies in terms of improper vendor selection, lengthening of the procurement process, incurring higher risks in operations and lack of transparency in the bidding process. Current reverse auction systems are largely theoretical or only focused on price bidding, and lack the technologies for real-time bidding, AI in vendor evaluation, fraud detection, analytics and security audits.

References

- [1] X. Chen, Y. Wang, and X. Zhao, **“Incentive-Based Reverse Auctions for Transportation Service Procurement with Cost Uncertainty and Delivery Risk,”** *PLOS ONE*, vol. 13, no. 11, 2018.
- [2] S. D. Jap, **“Online Reverse Auctions: Issues, Themes, and Prospects for the Future,”** *Journal of the Academy of Marketing Science*, vol. 30, no. 4, pp. 506–525, 2002.
- [3] D. R. Beil, **“Procurement Auction Design: A Review,”** *Interfaces*, vol. 40, no. 1, pp. 15–30, 2010.
- [4] V. Krishna, *Auction Theory*, 2nd ed. Academic Press, 2010.
- [5] J. Chai, J. N. K. Liu, and E. W. T. Ngai, **“Application of Decision-Making Techniques in Supplier Selection: A Systematic Review,”** *Expert Systems with Applications*, vol. 40, no. 10, pp. 3872–3885, 2013.
- [6] R. Carbonneau, K. Laframboise, and R. Vahidov, **“Application of Machine Learning Techniques for Supply Chain Demand Forecasting,”** *European Journal of Operational Research*, vol. 184, no. 3, pp. 1140–1154, 2008.
- [7] T. Wuest, D. Weimer, C. Irgens, and K.-D. Thoben, **“Machine Learning in Manufacturing: Advantages, Challenges, and Applications,”** *Production & Manufacturing Research*, vol. 4, no. 1, pp. 23–45, 2016.
- [8] I. Ahmed, A. Mahmood, and J. Hu, **“A Survey of Network Anomaly Detection Techniques,”** *Journal of Network and Computer Applications*, vol. 60, pp. 19–31, 2016.
- [9] M. Li, X. Zhang, and Y. Chen, **“Artificial Intelligence in Smart Procurement Systems: A Review,”** *IEEE Access*, vol. 8, pp. 123456–123470, 2020.
- [10] M. Pournader, Y. Shi, S. Seuring, and S. C. Koh, **“Blockchain Applications in Supply Chains, Transport and Logistics: A Systematic Review,”** *International Journal of Production Research*, vol. 58, no. 7, pp. 2063–2081, 2020.

Thank you